

SMART INDIA HACKATHON 2026

TITLE PAGE



SMART INDIA
HACKATHON
2026

Problem Statement ID – SIH26170

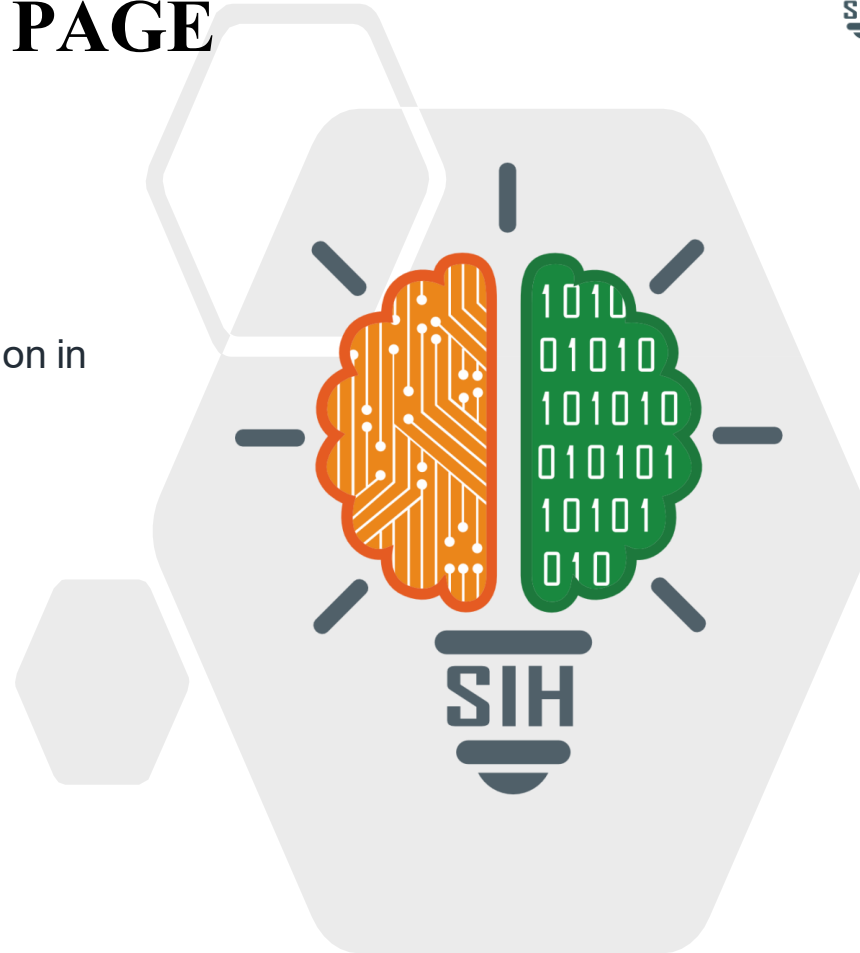
Problem Statement Title – AI-Driven Anomaly Detection in
Component Burn-In & Screening

Theme – Smart Automation

PS Category – Software

Team ID – NMIET Internal Ref: SIH2026-0076

Team Name – Tech Titans





BURNINSIGHT AI

Predictive Burn-In Screening



PROBLEM

- Static screening mainly checks whether a value crosses a fixed limit.
- A component can stay inside the limit while its behavior becomes abnormal.
- Subtle drift may only become obvious late in the burn-in cycle.
- Engineers need a way to prioritize suspicious components earlier.

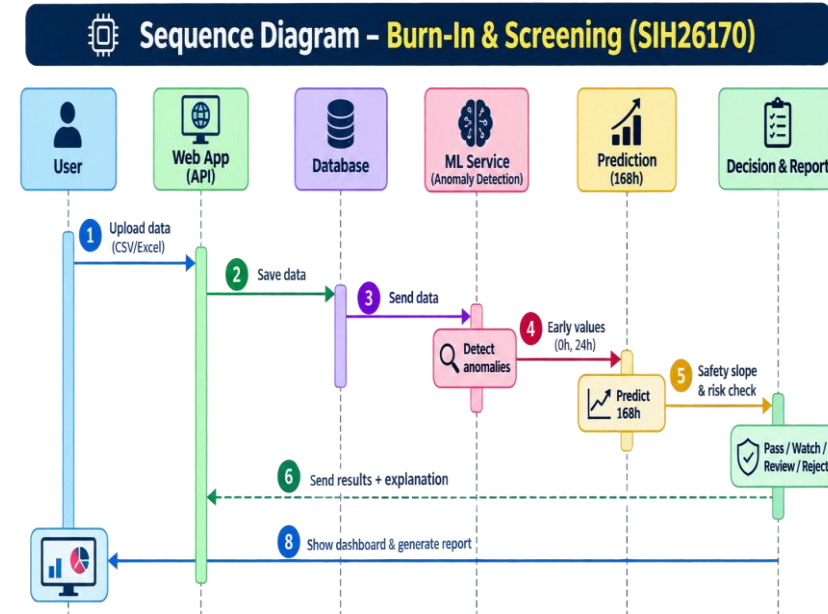
Key question:

“Is it within limit — and is it behaving normally?”

OUR SOLUTION

- Reads burn-in measurements from CSV / Excel.
- Finds lot-relative outliers using robust statistics + Isolation Forest.
- Uses early measurements (0h + 24h) to estimate 168h behavior.
- Combines anomaly, drift and configurable limits into a risk decision.
- Explains the decision with measurable reasons and an audit trail.

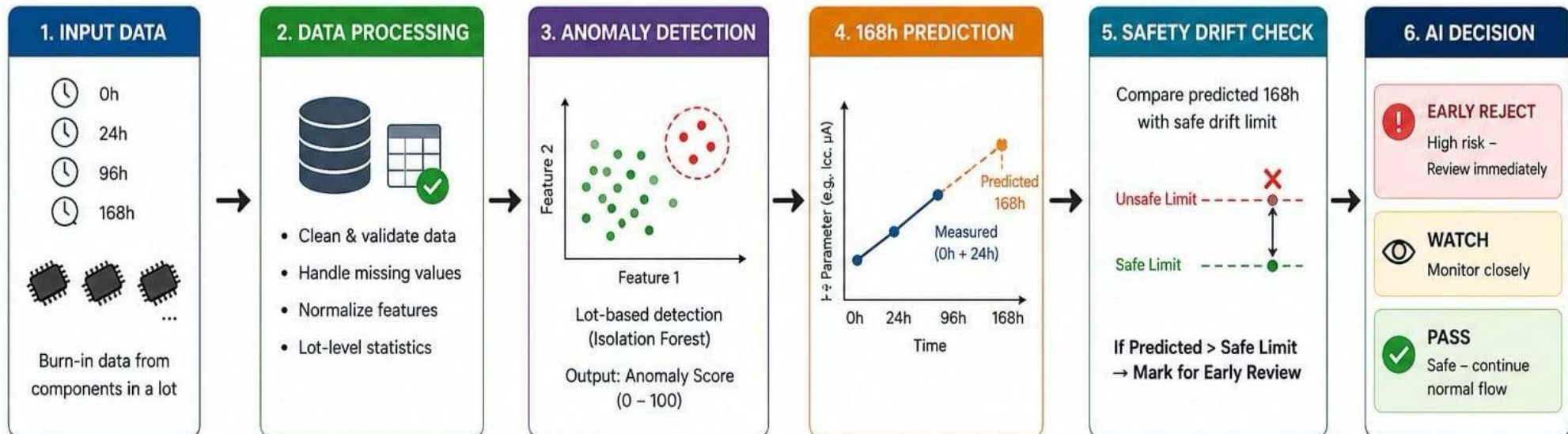
PASS • WATCH • REVIEW •
EARLY REJECT





TECHNICAL APPROACH

END-TO-END TECHNICAL PIPELINE



TECHNOLOGIES USED



Python



Pandas
NumPy



Scikit-learn
(Isolation Forest)



XGBoost /
Regression



SHAP



FastAPI
(Backend)



React
(Dashboard)



MySQL
(Database)



Cloud Storage
(For Data)

WHAT WE ACHIEVE

- ✓ Detect hidden defects early
- ✓ Predict future (168h) behavior
- ✓ Reduce false rejects & missed defects
- ✓ Improve reliability & save cost



FEASIBILITY AND VIABILITY



SMART INDIA
HACKATHON
2026



OPERATIONAL FEASIBILITY



Easy to use
web dashboard



Works for
engineers & operators



TECHNICAL FEASIBILITY



Uses pre-trained
ML models



Easy integration
with existing systems



VIABILITY



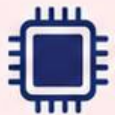
Early defect
detection



Reduces cost
and improves
reliability



CHALLENGES



Sensitive burn-in
data



Hidden anomalies
are hard to detect



Burn-in
Data



AI MODEL



Pass / Watch /
Review / Reject

LEARNS PATTERNS → DETECTS ANOMALIES → PREDICTS FAILURES



IMPACT / FEASIBILITY THROUGH AI



Higher product reliability



Lower cost and less
manual effort



Supports Make in India
& Industry 4.0



IMPACT AND BENEFITS



KEY BENEFITS



Early defect detection
Reduces field failures



Higher product reliability
Better quality and customer trust



Cost and time savings
Less rework and testing effort



Data-driven decisions
Focus on high-risk components



FUTURE SCOPE



Smarter AI models
with more data



Real-time monitoring
with live systems



Wider applications
across industries



Global standards
and collaborations



SCALABLE
SOLUTION



IMPACTS



Improved reliability
Predicts failures before they occur



Reduced field failures
Higher customer satisfaction



Cost-efficient testing
Saves time and resources



Supports Make in India
Stronger electronics manufacturing



BUSINESS MODEL



B2B Solution
For manufacturers
and test houses



**Subscription
Model**
Monthly/annual
plans



Easy Integration
Works with
existing systems



ADVANTAGES

Early
Detection



Higher
Accuracy



Lower
Cost



Real-time
Insights



**Our
Solution**



Better
Reliability



FUTURE IMPACT



Builds a data-driven quality ecosystem



Enables intelligent and self-reliant manufacturing



Contributes to Industry 4.0 and Atmanirbhar Bharat

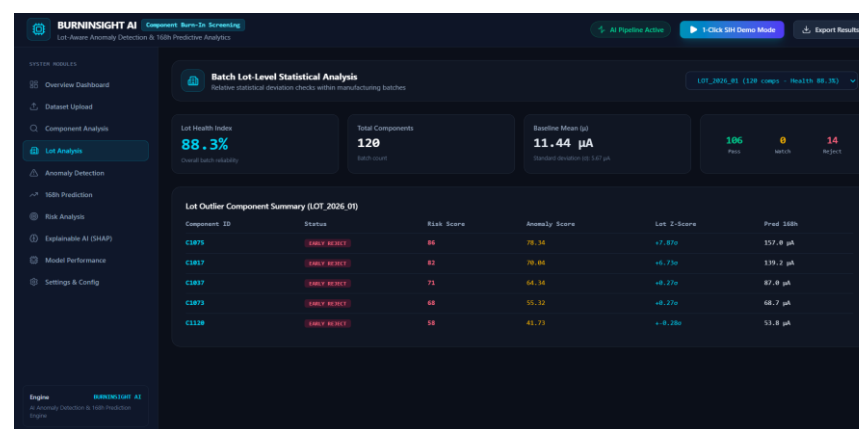
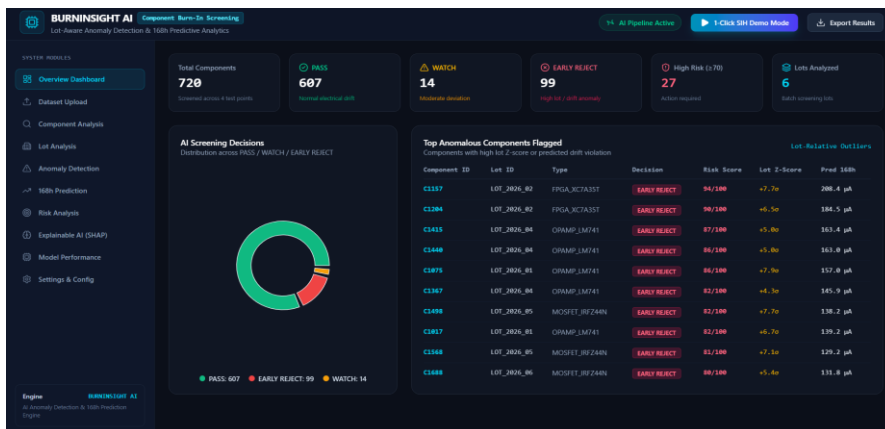


Sets a benchmark for AI-based testing solutions



RESEARCH AND REFERENCES

- ❖ NIST, *Nondestructive Tests Used to Insure the Integrity of Semiconductor Devices*, NBS SP 40059.
- ❖ Y. S. Obeng et al., *Towards Understanding Early Failures Behavior During Device Burn-In*, ECS J. Solid State Sci. Technol., 2016.
- ❖ Z. Chbili et al., *Modeling Early Breakdown Failures of Gate Oxide in SiC Power MOSFETs*, IEEE TED, 2016.
- ❖ Scikit-learn, Isolation Forest / Outlier Detection Documentation.
- ❖ SHAP, Explainable Machine Learning Documentation.



Burninsight-ai